Tutorial 07

Feedback paths tracking errors and stability

Preparation: Lecture 7. Close a negative-feedback loop, identify disturbance locations, calculate steady-state errors and interpret a stability boundary.

Submission credit: 1% of the course grade for submitted work; correctness is not marked.

Q1 derives feedback paths; Q2 applies given transfer functions. The ramp-tracking extension is optional.

Question 1 A flywheel speed loop under load

A frictionless flywheel has inertia $J = 0.4\ \text{kg·m}^2$. The reference r and actual speed ω are in rad/s; the controller applies torque $T = K(r - \omega)$, with $K = 1.6\ \text{N·m·s/rad}$. The positive opposing load d is in N·m. Use $J\,d\omega/dt = T - d$. R(s), Ω(s) and D(s) are the Laplace transforms of r, ω and d; assume zero initial conditions for transfer functions.

a) Draw the speed-feedback block diagram. Show where the opposing load enters and mark the summing signs. Derive Ω/R and Ω/D with the other input set to zero.

b) In two separate steady tests, hold r = 4 rad/s: first d = 0, then d = 0.8 N·m. Find the final speed, tracking error and controller torque in each case. At startup from rest with d = 0, check the initial torque against an 8 N·m positive limit.

MATLAB verification. Simulate both loads from rest with r = 4 rad/s applied at t = 0. Submit one figure with speed and torque in separate panels. Distinguish the loads with colours and line styles; include the 8 N·m limit. Report final values and initial torque. Base MATLAB is sufficient.

Question 2 Output disturbance and sensor bias

Use the temperature-deviation diagram and given closed-loop transfer functions below. R is the target temperature deviation, Y actual deviation, D an output disturbance and N sensor error (all in °C). U is heater-power deviation (W); K = 3 W/°C and G(s) = 2/(30s + 1) °C/W. All signals are deviations about an operating point. True tracking error is R − Y; measured controller error is E = R − Y − N. Assume zero initial conditions.

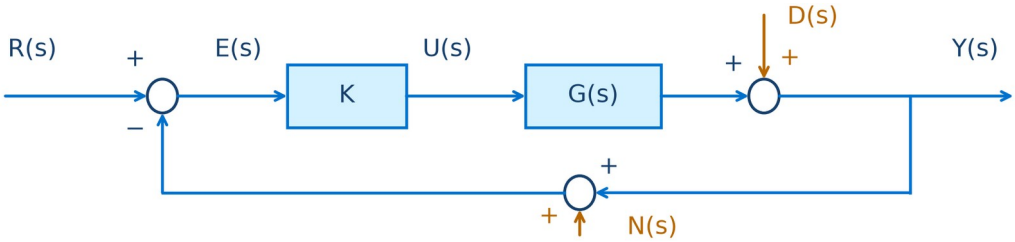


Figure 1. Temperature loop with output disturbance and sensor error.

T is complementary sensitivity and S is sensitivity. Set the other external inputs to zero in each test.

$$T = 6/(30s + 7)$$

$$S = (30s + 1)/(30s + 7)$$

$$Y/R = T; \qquad Y/D = S; \qquad Y/N = -T$$

a) Set R = 0. Evaluate two separate constant tests: (i) D = +2°C, N = 0; (ii) D = 0, N = +0.2°C. Use the supplied transfer functions to find final actual output, true tracking error and measured controller error. Explain the sign of the sensor-bias result.

b) As frequency increases without bound, use the given paths to find the limits of S and T and state which effect reaches Y: an output disturbance or sensor noise.

Question 3 Read a feedback stability boundary

A unity negative-feedback loop has L(s) = K/(s + 1)³, K > 0. All open-loop poles are −1 s⁻¹. **Figure 2** shows its Bode plots; ω is in rad/s. Use the plots and supplied closed-loop poles for the following checks; a cubic solution or full Nyquist test is not required.

a) Read the frequency where phase is −180°. Read the magnitude there for K = 4, 8 and 12. Which curve reaches the critical condition L(jω) = −1?

b) Use the supplied poles to classify each closed loop as stable, on the oscillatory boundary, or unstable. For K = 4, will its transient oscillations decay? For K = 12, will they grow?

$K = 4: \quad -2.587, \quad -0.206 \pm j1.375 \ s^{-1}$

$K = 8: \quad -3, \quad \pm j1.732 \ s^{-1}$

$K = 12: \quad -3.289, \quad +0.145 \pm j1.983 \ s^{-1}$

c) Explain why the negative-feedback sign alone is insufficient. Is the K = 8 case asymptotically stable?

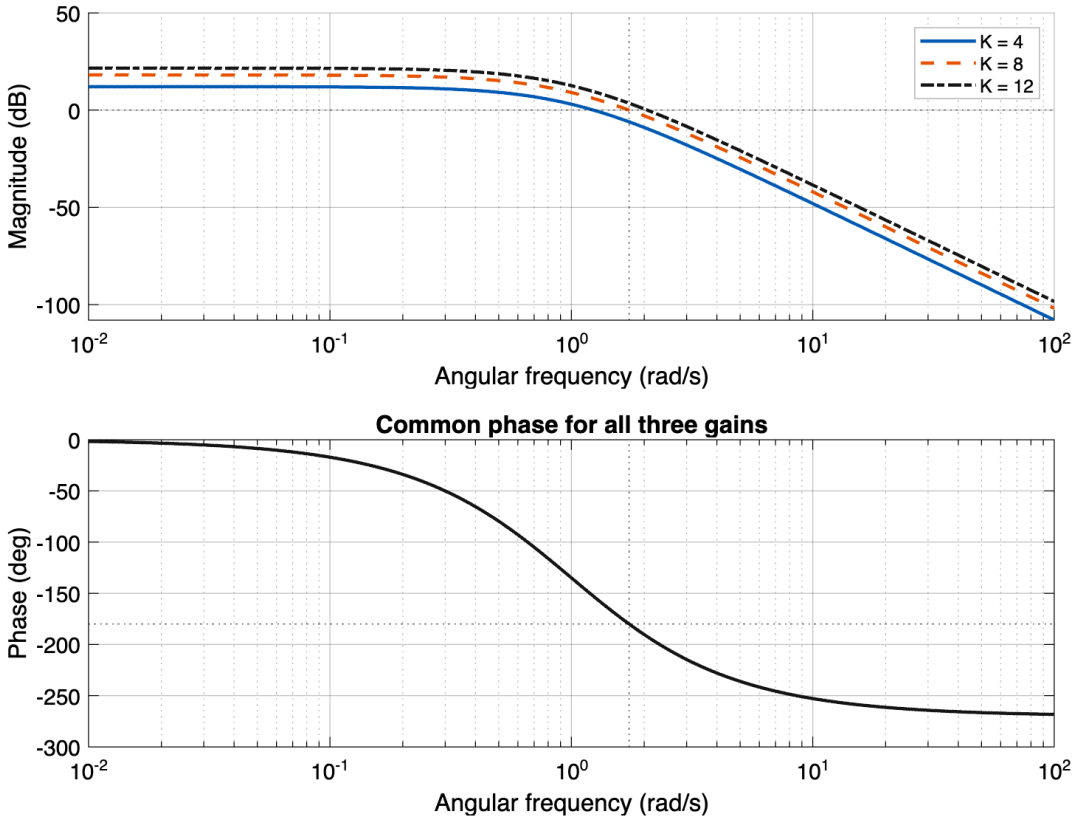


Figure 2. Open-loop Bode responses. The vertical dotted line marks the common −180° phase crossing; the horizontal dotted line marks 0 dB.

Extension One ramp tracking calculation

Optional. A stable unity-feedback loop has L(s) = 6/[s(s + 2)]. With zero initial conditions and no disturbance or noise, the reference is a unit ramp r(t) = t. Find the final tracking error using E = R/(1 + L) and the final-value theorem. Explain what the error represents.

Method reference

Lecture 7: feedback paths (slides 4–5), pole and transient interpretation (6–10), stability boundary (12–15), system type and tracking error (17–20), load and noise effects (22–27).

Ogata, Modern Control Engineering, 5th ed.: Sec. 5–8, pp. 225–227 (system type and tracking error); Problem A-5-23, p. 259 (rotor load disturbance); Sec. 8–6, p. 593 (separate reference, plant-input load and noise paths). Q1 adapts the resisting-load sign explicitly; Q2 places D after the plant. Q3 uses the Lecture 7 three-lag example. Printed page numbers are used.

Reference/tutorials/Tute 6/Tutorial 6 Questions, Q3–Q4: unity-feedback steady-state error. The original Tutorial 7 frequency-response tasks support the use of graph-reading practice.